🧮 Grade 12 Algebra – Assignment Questions

1. Solve the equation

$3(2x-5)-4(x+1)=2x+7$

2. Simplify the expression

$\frac{4x^2-9}{2x-3}$

and state any restrictions on x.

3. Solve the inequality

$\frac{x+2}{x-3}>1$

and express the solution set on a number line.

4. The function $f(x)=2x^2-3x+5$.

Find:

1. $f(2)$
2. The axis of symmetry
3. The minimum or maximum value.

5. Solve the system of equations:

$\left\{ \, \begin{array}{l}\textstyle 2x+y=7\\ \textstyle 3x-2y=4\end{array}\right.$

6. A geometric sequence has first term $a=6$ and common ratio $r=\frac{1}{2}$.

Find the **8th term** and the **sum of the first 10 terms**.

7. Factor completely:

x^3-4x^2-x+4

8. The quadratic equation

$5x^2-7x+k=0$

has **equal roots**.
Find the value of k.

9. A function is defined as

$f(x)=\frac{3x-1}{x+4}$

Find the **inverse function** $f^{-1}(x)$.

10. A company models its profit using the function

$P(x)=-2x^2+40x-96$

where x is the number of products produced (in hundreds).
Find:

1. The number of products that maximizes profit
2. The maximum profit.

If you want, I can also prepare:

- a **memorandum/answer key**,
- **another set of 10 or 20 questions**,
- or **mixed algebra + functions + calculus** questions for deeper practice.

Just tell me what you want next.